KEVIN M. FEDORUK

(949) 573-3015 | kevinmfedoruk@gmail.com | linkedin.com/in/kevinfedoruk | Portfolio | Orange County, CA | Open to Relocation | U.S. Citizen

PROFESSIONAL SUMMARY

Mechanical engineer seeking an entry-level design, propulsion, or test engineering role in aerospace and defense. B.S. in Mechanical Engineering (Propulsion concentration) from Embry-Riddle Aeronautical University and an MBA from Monmouth University, with hands-on experience across the full design cycle, including structural and thermal analysis, FEA, CAD modeling with GD&T, and instrumented testing on fabricated hardware. U.S. citizen able to obtain a Secret clearance. Proven leadership under pressure through six years of NCAA Division I / NAIA athletics.

TECHNICAL SKILLS

Software: SOLIDWORKS, MATLAB, ANSYS Workbench, LabVIEW, Microsoft Excel/Office
Engineering: Structural Analysis (Free-Body Diagrams, Hand Calculations), Fatigue & Damage Tolerance (DE-Goodman), Finite Element Analysis, CAD Modeling & GD&T, Mechanical & Machine Design, Mechanism/Actuation Design, AGMA Gear Stress Analysis, Shaft & Bearing Selection, Propulsion Systems (Turbojet, Turbofan, Ramjet, Rocket), Thermodynamics, Fluid Mechanics
Manufacturing & Testing: CNC Machining, MIG/TIG Welding, 3D Printing, Pressure Transducers, Thermocouples, Strain Gages
Languages: English, Spanish (Seal of Biliteracy)

ENGINEERING PROJECT EXPERIENCE

Twin Engine Thrust Vectoring Module | Personal SolidWorks Project *Sep. 2026 – Present*

- Designed a twin engine thrust vectoring module in SolidWorks from two 12 primary and 12 secondary petal variable area exhaust nozzles on a shared bulkhead, each in a printed two axis cardan gimbal (±15° pitch and yaw) actuated through servo arms and ball jointed pushrods, with a guided axial actuation ring for independent exit area control.
- Iteratively redesigned the primary petal hinge/crank geometry to reduce mechanism packaging while preserving actuation kinematics, and verified in CAD a 3.85 mm actuator stroke with interference-free operation from 0° to 15.5° petal rotation.
- Sized the connecting rods between the actuation ring and both the primary and secondary petals so they travel together and open to the target exit area, and packaged the secondary hinges within the nozzle envelope.
- Added the thrust vectoring gimbal around the finished nozzle and designed friction clamp stand in arms to stand in for servos; completed a GD&T drawing of the housing and am 3D printing the assembly to test fit and kinematics.

Senior Capstone – Rapid Lunar Transport Vehicle (RLTV) | Team Lead, 6-Person Team *Aug. 2023 – May 2024*

- Led a six-person team ("LUNA EV") through a NASA Artemis-aligned capstone; after helping design the full two-wheeled RLTV in preliminary design, focused detail design on a thermal protection system for its 250 W electric hub motors.
- Selected a 44°C-melt paraffin wax over a 54.67°C alternative through empirical testing, sized it via first-principles heat-transfer calculations (0.915 kg / 1.10 L required vs. 2.10 L capacity), and validated melt behavior with an ANSYS transient thermal simulation.
- Designed, fabricated, and instrumented a custom test article and rig, with a TIG-welded aluminum shell, brazed 16-compartment wax housing, dual-probe temperature logging (including a thermocouple in the motor core), and a digital torque sensor, to validate against a 30-minute, -10°C to 50°C requirement.
- Measured 73.4 W of real motor heat dissipation during testing; the system met the required 30-minute limit, and extended testing informed a recommendation for a lower-melting-point wax to add more margin, delivering the project $565.89 under budget.

Multi-Stage Gearbox — Machine Design Project | 3-Person Team *Aug. 2023 – Dec. 2023*

- Designed a four-gear, three-shaft gearbox to lift a 50-lb load over a required 10^8-cycle life, performing AGMA gear stress and DE-Goodman shaft fatigue analysis by hand, with every gear and shaft passing (min. 30.97 bending, 3.62 contact, 3.78 shaft).
- Modeled the complete gear train and housing in SOLIDWORKS with GD&T, sourcing hardware and evaluating manufacturability and cost tradeoffs to finalize a design that beat both the size and $400 budget constraints, delivered at $331.31.

MBA Marketplace Business Strategy Simulation | Team President, 9-Week Simulation *Jan. 2026 – May 2026*

- Directed a nine-week global business simulation, managing product development, pricing, production planning, advertising, and market expansion across five customer segments, three geographic regions, and multiple metropolitan markets.
- Finished 1st place in class and top 10% worldwide among more than 1 million Marketplace simulation participants.

EDUCATION

Embry-Riddle Aeronautical University — B.S., Mechanical Engineering, Propulsion *Aug. 2020 – May 2024*
GPA 3.602/4.000, Cum Laude. Relevant Coursework: Dynamics, Solid Mechanics, Machine Design, Computer-Aided Design, Heat Transfer, Thermodynamics, Fluid Mechanics, Airbreathing & Rocket Propulsion, Advanced Space Propulsion, Machine Shop, Electrical Engineering

Monmouth University — Master of Business Administration *Sep. 2025 – Aug. 2026*
GPA 3.910/4.000. Relevant Coursework: Managerial Economics, Managing & Leading, Project Management, Strategic Management

University of California, Riverside — MBA Coursework Completed *Sep. 2024 – Jun. 2025*

LEADERSHIP EXPERIENCE

MBA Marketplace Simulation Team President — Monmouth University *Jan. 2026 – May 2026*
Senior Capstone Team Lead — Embry-Riddle Aeronautical University *Aug. 2023 – May 2024*
Student-Athlete Advisory Committee (SAAC) Representative — Embry-Riddle Aeronautical University *Aug. 2020 – May 2024*

COLLEGIATE ATHLETICS

Six years of NCAA Division I / NAIA collegiate baseball — Monmouth University (2025–2026), University of California, Riverside (2024–2025), Embry-Riddle Aeronautical University (2020–2024) — while completing rigorous engineering and MBA coursework, developing leadership, communication, time management, resilience, teamwork, and the ability to perform under pressure.

AWARDS & SCHOLARSHIPS

Order of the Engineer (Inducted Member, May 2024) • Monmouth University Full Athletic Scholarship (2025–2026) • University of California, Riverside Academic Scholarship (2024–2025) • Embry-Riddle Aeronautical University Presidential Scholarship (2020–2024) • Embry-Riddle Aeronautical University Athletic Scholarship (2020–2024) • Embry-Riddle Aeronautical University Dean's List (2020–2024)